Project Assignment in 42114 Integer Programming

Technical University of Denmark – Fall 2026

Introduction

The project period officially starts on **September 24th** and the report must be submitted at the latest **November 9th at 23:59**. The questions to be answered are clearly marked in "question boxes". Please read the text and the questions carefully before answering them.

Info: Your final report must be uploaded to the appropriate group assignment on DTU Learn of the course. The report is expected to be not more than 6 pages long **excluding** tables, figures and appendices and must be written in Danish or in English. The project may be solved in groups and permissible group sizes are 2, 3, or 4 persons.

Modelling

A hospital plans elective surgeries over the next n days. Let S denote the set of surgeries and $D = \{1, \ldots, n\}$ the set of planning days.

Each surgery $s \in S$:

- requires t_s hours of operating-room time,
- has a preferred latest completion day r_s,
- incurs a penalty p_s for each day it is performed after its preferred completion day.

Each day $d \in D$ has H_d hours of operating-room capacity available, and every surgery must be scheduled exactly once during the planning horizon.

Answer the following questions:

Question 1

Formulate a Mixed Integer Linear Programming model that can be used to determine the surgery schedule that minimizes the total delay penalty. Clearly define all decision variables, constraints, and the objective function. If you use a Big-M construction, state an appropriate value for each Big-M parameter.

Question 2

Each surgery requires a specific surgeon. Let K denote the set of surgeons and $S_k \subset S$ the set of surgeries that require surgeon $k \in K$. A surgeon is considered to be working on a given day if they perform at least one surgery on that day. To reduce fatigue and maintain quality of care, the hospital has introduced a policy stating that no surgeon may work on three consecutive days. What changes are necessary to the model in Question 1 to enforce this requirement? Formulate all additional decision variables and constraints required.

Question 3

The hospital wishes to reserve capacity for emergency admissions. To achieve this, at least ρ hours of operating-room capacity must remain unused on at least 80% of the days in the planning horizon. What changes are necessary to the model in Question 1 to enforce this requirement? Formulate all additional decision variables and constraints required.

Question 4

The hospital has access to a robotic surgery system. A subset of surgeries, denoted by $R \subseteq S$, requires the use of this system. Operating the robot requires a dedicated technical team and a lengthy setup procedure. If at least one robotic surgery is performed on a given day, the robotic system must be activated for that day. Activating the robot incurs a fixed cost of $\gamma > 0$, independent of the number of robotic surgeries performed on that day. The hospital would like to jointly determine the surgery schedule and the robot activation schedule so as to minimize the sum of delay penalties and robot activation costs. What changes are necessary to the model in Question 1 to incorporate the robotic surgery system requirement? Formulate all additional decision variables, objective function modifications, and constraints required.

Lagrangian Relaxation

A technology company is evaluating 15 potential innovation initiatives. Each initiative requires resources and generates a strategic benefit if undertaken. The company has three development teams with limited capacity, and each selected initiative must be assigned to one of these teams. Team capacities are expressed in standardized effort units and are, respectively, 28, 32, and 35.
Some initiatives depend on the completion of others, meaning that certain projects can only be selected if their prerequisite projects are also chosen. In addition, some initiatives represent competing technological alternatives, so selecting one may preclude the selection of another. Management has also imposed strategic requirements that require an exact number of initiatives to be selected from specific categories. In particular, initiatives 2, 3, and 4 each require initiative 1; initiatives 7 and 8 require initiative 5; initiatives 9 and 10 require initiative 6; initiative 11 requires initiative 7; initiative 12 requires initiative 8; and initiative 15 requires initiative 10. Furthermore, initiatives 11 and 12 are mutually exclusive, meaning that at most one of the two initiatives may be selected. Finally, management requires that exactly three initiatives be selected from the set $\{2, 3, 4, 7, 8\}$.

The objective is to determine which initiatives should be undertaken and assigned to the three teams in order to maximize total strategic benefit while satisfying capacity limitations, project dependencies, the exclusivity requirement, and strategic selection targets.

Table 1: Initiative Data

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Strategic Benefit	12	24	20	28	10	11	30	27	22	25	40	38	32	34	36
Resource Consumption	14	8	7	10	13	12	9	8	7	8	12	11	10	9	10

Answer the following questions:

Question 5

Formulate the initiative selection problem as a binary integer programming problem. What well-known problem does this closely resemble? Justify your answer.

Question 6

You would like to investigate whether Lagrangian Relaxation can provide a tighter upper bound on the optimal objective than that provided by the LP relaxation of the IP model. Formulate a Lagrangian Relaxation of the problem, clearly indicating the constraints you relax.

Question 7

For the Lagrangian Relaxation you have identified, define its Lagrangian Dual Problem.

Question 8

Assuming that you use a subgradient method to update all Lagrangian Multipliers. What is the best upper bound you can obtain? How does it compare to the optimal objective value of the LP relaxation? Explain your observation and attach the code you use to generate this bound. Note that you will need to solve the LP relaxation of the problem given in Question 5 to answer this.

Question 9

Does your Lagrangian Relaxation terminate with a feasible solution to the IP model? If not, describe a Lagrangian Heuristic that can be used to convert this infeasible solution into a feasible one and demonstrate how it works. If your Lagrangian Relaxation terminates with a feasible solution, then demonstrate your heuristic on an infeasible solution to the original problem that is obtained during an iteration of the subgradient method.

Formulations

A security company is responsible for patrolling ten critical locations during a night shift. A collection of feasible patrol routes has already been generated. Each route visits a subset of locations and has an associated operating cost. Ideally, every location should be inspected exactly once during the shift. However, since the routes have been generated in advance, selecting multiple routes may result in some locations being inspected more than once. Although additional inspections are not prohibited, they are undesirable because they consume personnel resources, create unnecessary reporting work, and reduce the effectiveness of the patrol plan in other areas. For each location i, an over-coverage penalty of ρ_i is incurred for every inspection beyond the first. The objective is to determine the set of patrol routes that minimizes the sum of route operating costs and over-coverage penalties. The available routes and their associated costs are given in Table 2, while the over-coverage penalties are given in Table 3. Each location must be visited at least once in any feasible solution.

Answer the following questions:

Question 10

Devise a greedy heuristic to find a feasible solution to the problem. Clearly state what greedy principle you follow and report the solution you find together with its objective function value.

Table 2: Route information

Route	Cost	Locations visited
S_1	9	$\{1, 2, 5\}$
S_2	8	$\{1, 3, 4\}$
S_3	7	$\{2, 6\}$
S_4	6	$\{3, 7\}$
S_5	10	$\{4, 5, 8\}$
S_6	11	$\{6, 9\}$
S_7	12	$\{7, 10\}$
S_8	14	$\{8, 9, 10\}$
S_9	13	$\{1, 6, 8\}$
S_{10}	9	$\{2, 4, 7\}$
S_{11}	10	$\{1, 5, 9\}$
S_{12}	8	$\{6, 10\}$

Table 3: Over-coverage penalties

Location i	1	2	3	4	5	6	7	8	9	10
Penalty ρ_i	2	1	1	3	1	2	2	1	3	2

Question 11

Formulate a Mixed-Integer Linear Program that can be used to find an optimal solution to the problem. Find the optimal solution to its LP relaxation. Within what interval must the optimal objective value lie?

Question 12

Using the optimal solution you find to the LP relaxation, evaluate two nodes of the Branch-and-Bound tree. Clearly identify the variables you branch on, the resulting solution to each node, and their associated objective function values. Based on the partial Branch-and-Bound tree, what can you now say about the interval within which the optimal objective function value lies?

Question 13

Identify a valid relaxation of the problem that has a single constraint, and which cannot be obtained by simply removing constraints from your formulation in Question 11. What bound does this relaxation provide?

Question 14

Without solving an LP relaxation identify a valid dual bound for the security patrol problem. Indicate clearly how you derive this bound.

Dynamic Programming

Following a natural disaster, a humanitarian organization is preparing a shipment of supplies for a temporary field hospital. Four types of equipment are required for the mission:

1. Diagnostic equipment,
2. Surgical equipment,
3. Patient monitoring equipment,
4. Communication equipment.

For each type of equipment, three alternative packages are available. Packages differ in terms of the medical benefit they provide and the amount of cargo space they require. To ensure that all critical functions of the field hospital are covered, exactly one package must be selected for each type of equipment.

The shipment will be transported by a small aircraft with limited cargo capacity. The combined cargo space required by the selected packages may not exceed 13 units. The aim is to select the equipment packages that maximize the total medical benefit provided by the shipment while respecting the aircraft's cargo capacity. All the necessary parameters are given in Table 4.

Table 4: Available equipment packages

Item	Group	Profit	Weight
A1	Diagnostic Equipment	10	2
A2	Diagnostic Equipment	14	3
A3	Diagnostic Equipment	20	5
B1	Surgical Equipment	8	2
B2	Surgical Equipment	15	4
B3	Surgical Equipment	18	5
C1	Monitoring Equipment	5	1
C2	Monitoring Equipment	12	3
C3	Monitoring Equipment	16	4
D1	Communication Equipment	7	2
D2	Communication Equipment	11	3
D3	Communication Equipment	19	5

You observe that dynamic programming can be used to identify the best set of equipment types to take.

Question 15

Identify the stages and states of your dynamic program.

Question 16

Give the forward recursion for the dynamic program that will compute the optimal set of equipment types, i.e., the one that gives maximum medical benefit.

Question 17

Solve the dynamic program. Report intermediate calculations in the form of a table containing the optimal objective value for each stage-state combination. Report the optimal solution and show how it can be obtained using your table.

Question 18

Formulate the optimization problem as a binary integer program and solve it using Julia/JuMP. Compare the solution obtained with the one obtained via dynamic programming.